1. The potential difference across all the resistors is equal to the EMF applied.
2. The current flows through resistors will differ according to the value of the resistor.

   The current flows through resistor R_1 is referred as I_1, whereas the current flows through resistorR_2, is referred as I_2.

$$I_1 = \frac{V}{R_1} \qquad I_2 = \frac{V}{R_2}$$

3. Hence, the sum of the currents flows through the individual resistor is equal to the flow of total current in the circuit. Using this, we can derive the formula for the total resistance *(R)*.

$$I = I_1 + I_2$$

As per ohm's law, $\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2}$

$$\frac{V}{R} = V\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

Cancelling 'V' on both sides,

$$\frac{1}{R} = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

Again simplifying the above formula by taking LCM,

$$\frac{1}{R} = \left(\frac{R_2 + R_1}{R_1 R_2}\right)$$

Cross multiplying: $R(R_1 + R_2) = R_1 \cdot R_2$

$$R = \frac{R_1 R_2}{R_1 + R_2}$$

4. There will be current flow in other paths, even if any of the paths is disconnected. It is shown in Figure 1.23

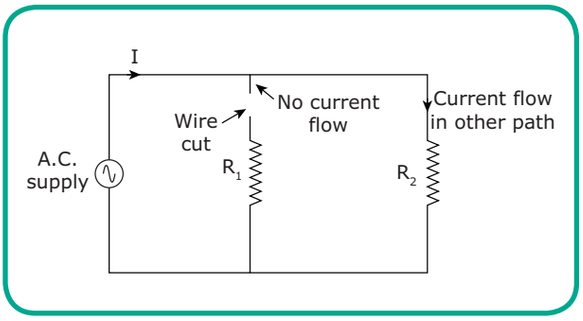


Figure 1.23

5. The current in this circuit takes two or more branch paths from main path. The current takes branch path (i.e. flows through each resistor) is called branch current.

1.6.1 Uses of Parallel Connection

At homes and industries, this type of connection is used.

Problems

1. Two resistors of values 10Ω and 15Ω are connected in parallel with 60V power supply. Prove that the sum of the branch currents is equal to the total current.

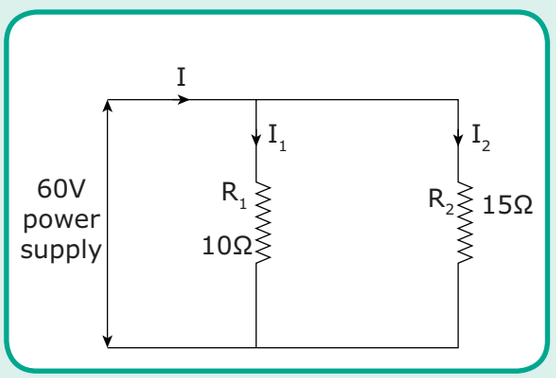


Figure 1.24

Total Resistance

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{10 \times 15}{10 + 15} = \frac{150}{25} = 6\Omega$$

$R = 6\ \Omega$

(continued)

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